

Artificial Intelligence in the Knowledge Economy

Capability, autonomy, organization, and the distribution of gains

Carlos Gómez Puican
Paper by Enrique Ide & Eduard Talamàs

Repository 5

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<https://github.com/carlosgomezp28/ai-05-ide>

1. Research question and contribution

Question

How does scalable AI that can acquire tacit knowledge reorganize knowledge work—and who gains when AI can act rather than merely advise?

Organizational margin

- ▶ Humans sort into routine production and exceptional problem solving.
- ▶ AI changes occupations, matching, spans of control, and wages in equilibrium.

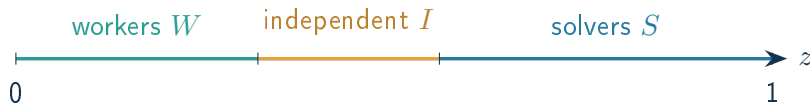
Two technological dimensions

- ▶ Capability: how difficult a problem AI can solve.
- ▶ Autonomy: whether AI can pursue work independently or only assist.

Central tradeoff: autonomous AI raises output more, but directs relatively more gains toward the top.

2. Before AI: a knowledge hierarchy

Humans have observable knowledge $z \in [0, 1]$, with CDF G and a continuous, strictly positive density.



- ▶ An independent producer spends one unit of time on one opportunity.
- ▶ Workers attempt routine cases; solvers handle the exceptions workers cannot solve.
- ▶ Knowledge is tacit: the worker must ask before learning whether the solver knows the answer.
- ▶ Equilibrium exhibits **management by exception**: $W \preceq I \preceq S$, with positive assortative matching.

3. The economic problem and team size

A problem has difficulty $x \sim U[0, 1]$. Knowledge z therefore has a direct interpretation:

$$\Pr(\text{solve independently} \mid z) = \Pr(x \leq z) = z.$$

A worker fails with probability $1 - z$. Each request consumes $h \in (0, 1)$ units of solver time, so full use of one solver requires

$$h n(z)(1 - z) = 1 \quad \implies \quad \boxed{n(z) = \frac{1}{h(1 - z)}}.$$

Firms choose the organization that maximizes expected output net of wages and compute rent. For a human solver s and human workers z ,

$$\Pi_2^{nA}(s, z) = n(z) [s - w(z)] - w(s), \quad z \leq s.$$

Competition imposes non-positive feasible profits, zero profit for used configurations, and market clearing.

4. AI technology: capability is not autonomy

One unit of compute creates an AI agent. Keep the two dimensions separate:

	Capability z_{AI}	Autonomy
Meaning	Knowledge cutoff: fraction of difficulties AI can solve	Feasible roles: whether AI may pursue production independently
Values	$z_{AI} \in [0, 1)$	Two paper benchmarks: autonomous or non-autonomous
Labels	Basic if $z_{AI} \in \text{int } W$; advanced if $z_{AI} \in \text{int } S$	Autonomous = co-worker and co-pilot; non-autonomous = co-pilot only

Controlled comparison

Proposition 6 holds z_{AI} fixed and changes autonomy. Proposition 5 varies z_{AI} within the autonomous regime.

5. Post-AI firms and equilibrium intuition

Configuration	Profit	Economic role
Independent human	$z - w(z)$	Human pursues one opportunity
Independent AI	$z_{\text{AI}} - r$	AI pursues an opportunity
Human team	$n(z)[s - w(z)] - w(s)$	Human workers, human solver
AI solver (top automated)	$n(z)[z_{\text{AI}} - w(z)] - r$	Human workers, AI co-pilot
AI workers (bottom automated)	$n(z_{\text{AI}})[s - r] - w(s)$	AI co-workers, human solver

- ▶ Autonomous AI can enter all three AI roles. Abundant compute guarantees some independent AI production, which pins down $r^* = z_{\text{AI}}$.
- ▶ Non-autonomous AI cannot be an independent producer or worker; only the top-automated configuration remains available.
- ▶ Scalability makes AI a large homogeneous supply of knowledge, so its effect is not the same as adding heterogeneous humans.

6. Propositions 1–4: setup for distributional results

Propositions 1–2: equilibrium

- ▶ Pre-AI equilibrium: unique, efficient, $W \preceq I \preceq S$, positive assortative matching.
- ▶ Under $h < h_0$, $I = \emptyset$.
- ▶ Autonomous-AI equilibrium: unique and efficient, with

$$W^* \preceq \{z_{\text{AI}}\} \preceq S^*.$$

- ▶ AI always does some independent production and $r^* = z_{\text{AI}}$.

Propositions 3–4: reorganization

- ▶ Basic AI: $W^* \subset W$, $S \subset S^*$; marginal humans move toward solving.
- ▶ Advanced AI: $W \subset W^*$, $S^* \subset S$; movement reverses.
- ▶ Continuing workers' productivity and continuing solvers' spans depend on how AI changes their matches.

Occupational displacement changes both productivity and the division of team output.

7. Proposition 5: who wins with autonomous AI?

Let w be the pre-AI wage and w^A the autonomous-AI wage. Define

$$B = \{z \in [0, z_{\text{AI}}] : w^A(z) > w(z)\}, \quad T = \{z \in [z_{\text{AI}}, 1] : w^A(z) > w(z)\}.$$

Proposition 5

Under $h < h_0$, competitive pre/post equilibria, $z_{\text{AI}} \in [0, 1)$, and compute abundance uniformly over admissible capabilities, there is

$$\boxed{\bar{z}_{\text{AI}} \in \text{int } W}, \quad \boxed{B \neq \emptyset \iff z_{\text{AI}} > \bar{z}_{\text{AI}}}, \quad \boxed{T \neq \emptyset \quad \forall z_{\text{AI}} \in [0, 1]}.$$

Winners lie at the extremes: $B = [0, z_b)$ and $T = (z_t, 1]$. The human with knowledge exactly z_{AI} loses because $w^A(z_{\text{AI}}) = z_{\text{AI}} < w(z_{\text{AI}})$ under $h < h_0$.

8. Why Proposition 5 holds: match versus share

Write the wage change as $\Delta(z; z_{\text{AI}}) = w^A(z; z_{\text{AI}}) - w(z)$. The threshold is constructed from

$$\Delta(0; \bar{z}_{\text{AI}}) = 0,$$

with $\Delta(0; z_{\text{AI}})$ continuous and strictly increasing in capability over the relevant region.

Bottom of the distribution

Match effect: basic AI can leave low- z workers with a less knowledgeable solver.

Share effect: the matched solver's outside option falls toward independent-production income, so workers retain more team output.

Bottom wins only when the share effect dominates: $z_{\text{AI}} > \bar{z}_{\text{AI}}$.

Top of the distribution

The same two forces can oppose one another, but under $h < h_0$ team sizes are large enough that the positive share effect dominates.

Hence $T \neq \emptyset$ for every admissible, non-superintelligent capability.

9. Critical interpretation: two thresholds, two questions

Autonomous AI: who wins at the bottom?

Non-autonomous AI: is the co-pilot adopted?

$$B \neq \emptyset \iff z_{\text{AI}} > \bar{z}_{\text{AI}}, \quad \bar{z}_{\text{AI}} \in \text{int } W.$$

Hold the technology autonomous and ask whether its **capability** is high enough for the share effect to beat the match effect.

$$z_{\text{AI}} > w(0).$$

Hold AI non-autonomous and ask whether its advice is valuable enough for any firm to use it as a solver.

The bottom-winner cutoff $\bar{z}_{\text{AI}}$ is not the adoption cutoff $w(0)$.

Capability and autonomy are separate inputs to the equilibrium, not competing labels for one scale.

10. Proposition 6: non-autonomous equilibrium and adoption

Keep capability z_{AI} and compute availability fixed, but prohibit AI from pursuing production opportunities. AI can only be a solver/co-pilot.

Equilibrium

Under admissible z_{AI} , abundant compute, $h < h_0$, and the corresponding pre-AI and autonomous equilibria, the non-autonomous equilibrium is unique, efficient, and labor-income maximizing. Because some compute is idle,

$$\boxed{r^N = 0}.$$

$z_{AI} \leq w(0) \implies W_a^N = \emptyset, \quad w^N = w, \quad \text{same human occupations;}$

$z_{AI} > w(0) \implies \text{AI is adopted as a solver by the least knowledgeable humans.}$

When adopted, human AI-users, human-only workers, and human solvers are nonempty; independent humans may be absent.

11. Proposition 6: output and wage incidence

Let Y^A, w^A denote autonomous outcomes and Y^N, w^N non-autonomous outcomes.

1. **Output:**

$$\boxed{Y^N < Y^A}.$$

Non-autonomy binds: compute that could produce independently is idle.

2. **Some loser relative to pre-AI:**

$$\exists z \in (0, 1] : w^N(z) \leq w(z),$$

strictly for some z when $z_{AI} > w(0)$.

3. **Bottom interval favors non-autonomy:**

$$\exists \varepsilon > 0, \forall z \in [0, \varepsilon) : w^N(z) \geq \max\{w(z), w^A(z)\},$$

strictly when AI is adopted.

4. **Top interval favors autonomy:**

$$\exists \varepsilon > 0, \forall z \in (1 - \varepsilon, 1] : w^N(z) \leq w^A(z),$$

strictly when $z \neq 1$.

12. Where I did not believe the AI

Proposition 5

Discrete check: when do low-knowledge workers gain from autonomous AI?

Consider a low-knowledge worker with knowledge l , initially matched with a human solver of knowledge $s \geq l$.

A worker fails with probability $1-l$, and each failed problem uses h units of solvertime. Define

$$q \equiv h(1-l) \quad n(l) = \frac{1}{q}$$

Let w_H be the wage of the human solver.

1. Before AI

Zero profit for a human team implies

$$\begin{aligned} n(l)[s - w_L] - w_H &= 0 \\ \frac{1}{q}[s - w_L] &= w_H \end{aligned}$$

so

$$w_L = s - q w_H$$

Thus the worker's wage depends positively on the solver's knowledge s , but the worker must indirectly compensate the solver through $q w_H$.

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2. Autonomous AI as solver

Now introduce autonomous AI with capability

$$a \equiv z_{AI}, \quad l < a < s$$

Because autonomous AI can also produce independently, its competitive rental rate is

$$r = a$$

If AI becomes the solver, zero profit gives

$$\begin{aligned} n(l)[a - w_L^A] - a &= 0 \\ \frac{1}{q}[a - w_L^A] - a &= 0 \end{aligned}$$

Therefore,

$$\frac{1}{q}[a - w_L^A] = a$$

and

$$w_L^A = a(1-q)$$

12. Hand derivation: wage effects and cutoff

3. Wage effect on the bottom worker

Define

$$\Delta_L = w_L^A - w_L$$

Then

$$\Delta_L = a(1-q) - (s - qW_H)$$

$$\Delta_L = \underbrace{(a-s)}_{\text{match effect}} + \underbrace{q(W_H-a)}_{\text{share effect}}$$

Since $a < s$

$$a-s < 0,$$

so replacing the human solver with AI worsens the worker's match quality

But, if $W_H > a$

$$q(W_H-a) > 0,$$

because AI is a cheaper solver. The worker therefore keeps a larger share of team output

Hence the worker gains only if

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share effect > match effect

4. Capability cutoff

Set

$$\Delta_L = 0,$$

Then

$$a-s+q(W_H-a)=0,$$

so

$$a(1-q) = s - qW_H$$

Therefore the cutoff is

$$\bar{a} = \frac{s - qW_H}{1-q}$$

Since $1-q > 0$

$$w_L^A > w_L \iff a > \bar{a}$$

So, holding autonomy fixed, low-knowledge workers benefit only when AI capability is sufficiently high

$$\Delta_L = \underbrace{(a-s)}_{\text{match effect}} + \underbrace{q(W_H-a)}_{\text{share effect}},$$

$$\Delta_L > 0 \iff a > \bar{a}.$$

12. Hand derivation: numerical check and verdict

5. Numerical check	
Take	$S = 0.8, W_H = 1, q = 0.4$
Then	$w_L = 0.8 - 0.4(1) = 0.4$
and	$\bar{a} = \frac{0.8 - 0.4}{0.6} = \frac{2}{3} \approx 0.667$
If $a = 0.60 < \bar{a}$,	$w_L^A = 0.60(0.60) = 0.36 < 0.4$ $\Rightarrow$ Bottom worker loses
If $a = 0.70 > \bar{a}$,	$w_L^A = 0.70(0.60) = 0.42 > 0.40$ $\Rightarrow$ Bottom worker gains
Verdict	Bottom gains $\Leftrightarrow Z_{AI} > \bar{Z}_{AI}$

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In this discrete analogue, the threshold comes from the positive share effect eventually dominating the negative match effect.	
The Key condition is about AI capability, not autonomy, because autonomy is held fixed throughout this comparison.	

Verdict

Discrete hand check: holding autonomy fixed, the low-knowledge worker gains iff AI capability crosses a cutoff. The positive share effect must dominate the negative match effect. This illustrates Proposition 5's mechanism; it is not a proof of the continuum result.

13. Lean bridge: Proposition 5

a. Original mathematical claim.

$$\exists \bar{z}_{AI} \in \text{int } W : \quad B \neq \emptyset \iff z_{AI} > \bar{z}_{AI}, \quad T \neq \emptyset \forall z_{AI} \in [0, 1).$$

```

-- Lean statement of Proposition 5 (omitted)
∃ threshold : ℝ,
  threshold ∈ interior (humanWorkers pre) ∧
  ∀ zAI, AdmissibleAI zAI →
    ∀ post,
      CompetitiveEquilibrium economy zAI AMode.autonomous post →
        ((bottomWinners zAI pre post).Nonempty ↔
          threshold < zAI) ∧
          (topWinners zAI pre post).Nonempty
theorem proposition5_main : proposition5Spec := by
  sorry

```

c. Plain-English mapping.

- Objects: pre/post equilibria, z_{AI} , threshold.
- Assumptions: compute abundance, admissibility, competitive equilibrium, $h < h_0$.
- Lean checks the statement type-checks; sorry means the analytical proof is not verified.

14. Formalization boundary and failed closeout

What succeeded	What remains
v11 model and competitive-equilibrium definition represented; six transparent Specs; build/interface check passed.	Six theorem bodies remain sorry: <code>proposition1_main</code> through <code>proposition6_main</code> ; continuum-equilibrium and comparative-static proofs remain open.

Proposition 5: proof obligation and fidelity

Open nontrivial proof step: construct `threshold` in `interior (humanWorkers pre)` and establish, for every admissible z_{AI} and autonomous competitive post equilibrium, the bottom-winner equivalence and top-winner nonemptiness.

Spec fidelity: source map and Spec match on $z_{AI} \in [0, 1)$, the cutoff in `int  $W$` , and both winner conclusions. Lean adds representational `Economy/outcome` binders and named wrappers for maintained uniform abundance, $h < h_0$, and competitive pre/autonomous-post equilibria; no extra-assumption boundary or domain change is recorded, and independent source audit remains pending.

Status and external blocker

Partially formalized: a successful build does not prove the six propositions. Closeout also awaits the documented external `AppliedModelingLib` engine-state prerequisite.

15. Capability, autonomy, and the final takeaway

	Capability z_{AI}	Autonomy
Policy question	How capable must AI be for a group to gain?	Which economic roles may AI perform?
Key threshold	Autonomous bottom winners iff $z_{AI} > \bar{z}_{AI}$, with $\bar{z}_{AI} \in \text{int } W$	Non-autonomous adoption iff $z_{AI} > w(0)$
Distribution	Basic versus advanced capability reverses occupational displacement	Bottom interval favors co-pilot-only AI; top interval favors autonomous AI
Output	Better capability changes which problems can be solved	Holding capability fixed, autonomy produces strictly more output than non-autonomy

Final economic takeaway

Capability determines what AI knows; autonomy determines the roles in which that knowledge can act. Holding capability fixed, autonomous AI produces more output than non-autonomous AI, but shifts relative gains toward the top.